L2.4 Limits at Infinity

Work in your group. Check your answers against the Quarto tonight.

Limits at infinity and horizontal asymptotes

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1. Evaluate. Divide top and bottom by the highest power of x in the **denominator**, and show the division.

$$\lim_{x \to \infty} \frac{4x - 3}{x^2 + 2}$$

2. Evaluate.

$$\lim_{x \to \infty} \frac{6x^2 - x}{2x^2 + 5}$$

> **Stop.** Which terms did the division kill, and why? Each one should be $\frac{a}{x^r}$.

The three asymptote rules, derived

3. Identify the horizontal or slant asymptote.

$$f(x) = \frac{x^3 - 2x^2 + 1}{-3x^3 + 2x - 1}$$

4. Identify the horizontal or slant asymptote.

$$g(x) = \frac{x + 1}{x^{99} + 1}$$

5. Identify the horizontal or slant asymptote.

$$h(x) = \frac{3x^6 + x^5}{2x^5}$$

> **Stop.** One of these three is not a horizontal asymptote at all. Which degree comparison put it in a different case?

Limits at $-\infty$

6. Evaluate.

$$\lim_{x\to\infty} \frac{\sqrt{9x^2+2}}{2x+5}$$

7. Same function, other direction.

$$\lim_{x\to-\infty} \frac{\sqrt{9x^2+2}}{2x+5}$$

Stop. Your two answers should differ. If they do not, look at what $\frac{1}{x}$ becomes when it goes inside the radical.

8. Sketch a graph of

$$f(x) = \frac{3x^2+4x+5}{\sqrt{16x^4-81}}$$

Find the vertical asymptotes and the horizontal asymptotes first. Say how many horizontal asymptotes there are, and why that is different from the last problem.

Slant asymptotes

9. Find the slant asymptote by polynomial long division.

$$y = \frac{2x^2 + 3x - 1}{x + 1}$$

> **Stop.** After the division, one term goes to 0. That term is the reason the line is an asymptote — point at it.

10. Evaluate.

$$\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - x \right)$$

Reading asymptotes backwards

11. Draw an **even** function which has the lines $y = 1$, $x = -4$ and $x = -1$ among its asymptotes.

> **Stop.** An even function is symmetric about the y-axis. What does that force about the rest of the picture?

12. BOARDS **Group work.** Let

$$P(x) = a_m x^m + \cdots + a_1 x + a_0 \qquad \text{and} \qquad Q(x) = b_n x^n + \cdots + b_1 x + b_0.$$

Your group gets one case. Find $\lim\limits_{x\to\infty} \dfrac{P(x)}{Q(x)}$, **derive it** from the limit at infinity theorem, and be ready to report out.

(a) $m = n$

(b) $m < n$

(c) $m > n$